

Tunelt

Complete System Guide, Under-the-Hood Pipeline & Live Benchmarks

Simple, plain-language breakdown of Tunelt playlist sequencing technology

1. What is Tunelt?

Tunelt is a smart music organizer built to solve a major pain point in digital music streaming: chaotic, randomly ordered playlists.

When you listen to a playlist on default shuffle, a calm acoustic guitar song might be immediately followed by a blasting heavy metal beat or aggressive track. These sudden energy jumps ruin your mood and break your focus.

Tunelt connects directly to your YouTube Music account, uses Artificial Intelligence (Google Gemini) to analyze the tempo and energy level of every song, and automatically reorders the tracks into a smooth, satisfying listening journey.

Simple Analogy: Tunelt is Your Personal AI DJ

Think of Tunelt as a human DJ sitting inside your phone. Instead of shuffling songs randomly, the DJ looks at every track in your queue, evaluates how fast and intense each song feels, pairs similar tracks together, and builds a smooth continuous mix.

2. What Problem Does Tunelt Solve?

Traditional music players pick songs randomly or by popularity, completely ignoring transition quality. Tunelt solves 3 common issues:

1. Sudden Mood & Volume Spikes: Prevents peaceful relaxation tracks from being ruined by high-decibel dance beats.
2. Loss of Focus During Work: Keeps song energy steady so you stay in an uninterrupted productive state while studying or coding.
3. Boring Workout Sequences: Gradually builds up song intensity step-by-step so your music peaks right when you reach your maximum exercise effort.

3. Under the Hood: Full Step-by-Step Pipeline

Here is the complete step-by-step process of what happens behind the scenes from input to final export:

Phase 1: Playlist Fetching & Connection

You log in securely with your Google account. Tunelt connects to the YouTube Music API to read your selected playlist, retrieving track titles, artist names, video IDs, and genre tags.

Phase 2: AI Sonic Intensity & Tempo Analysis

Tunelt sends track details to Google Gemini AI. The AI evaluates each song like a music expert, assigning two primary values:

- Beats Per Minute (BPM): The speed of the track (e.g., 80 BPM = slow chill; 130 BPM = fast dance).
- Sonic Intensity (0.0 to 1.0): How calm or intense the song feels (e.g., 0.2 = peaceful lo-fi; 0.85 = heavy workout beat).

Phase 3: Database Caching (Instant Speed)

To keep Tunelt fast and save network bandwidth, all song scores are stored in a database cache. If you or another user selects the same song in the future, Tunelt retrieves the saved scores instantly without querying the AI again.

Phase 4: Flow Engine Selection & Re-Ordering

You select your desired vibe (Build-Up Mode, Drift Mode, Unhinged Mode, or Frame Engine). Tunelt's algorithm engines reorder your track list according to your choice so every song transition flows naturally.

Phase 5: Visual Energy Graph & Audio Previews

Before exporting, Tunelt renders an interactive energy graph on your screen. You can visually inspect the playlist curve and click any song to listen to a 30-second audio preview.

Phase 6: Direct Export to YouTube Music

With one click, Tunelt creates a brand-new, perfectly ordered playlist directly inside your YouTube Music account or lets you download the sequence as a CSV file.

4. The 4 Flow Modes Explained Simply

Tunelt features 4 distinct listening engines, each tailored to different activities and moods:

Build-Up Mode (Rise Engine) - Best for: Workouts & Running

Starts with calm, easygoing songs and steadily climbs uphill like a staircase. Energy builds smoothly step-by-step, reaching an exciting peak right at the end of your session.

Drift Mode - Best for: Deep Work, Studying, Coding, & Reading

Filters out aggressive energy jumps and keeps the playlist energy flat and steady. Song transitions are gentle so your mind remains focused without distraction.

Unhinged Mode - Best for: High-Energy Gaming & Parties

Built for high excitement! Sequences high-intensity tracks back-to-back, occasionally dropping in surprise curveball tracks paired with musical key alignment so it sounds amazing.

Frame Engine - Best for: Cinematic Experience

Organized like a 3-Act movie: Act 1 acts as a gentle warm-up intro, Act 2 provides the main smooth groove, and Act 3 brings a relaxing wind-down at the end.

5. System Architecture (Surface Level)

Tunelt is engineered using modern, high-performance web components:

- Frontend (User Interface): Built with Next.js 16, React, and Framer Motion. Handles screen displays, interactive energy charts, and 30-second audio previews.
- Backend (Server & API): Built with Express.js and Node.js. Manages Google OAuth authentication, connects to YouTube Music, and runs the 4 flow sorting engines.
- AI Engine: Powered by Google Gemini 3.1 Flash Lite API for rapid music energy and mood evaluation.
- Database Cache: Uses Prisma DB to store song metrics for instant future reuse.

6. Combined Live Algorithm Benchmarks

To prove how effectively Tunelt sequences music, we executed live benchmark tests on both a standard 21-track user playlist and a high-scale 500-track dataset. Here are the combined real-world results:

Benchmark 1: Standard 21-Track Playlist Evaluation

Engine Mode	Tracks Retained	Smoothness Score	Jarring Jumps (>0.35)	Vibe Characteristics
Rise Algorithm	21 / 21 (100%)	96.4 / 100	0	Staircase Ascent
Frame Engine	21 / 21 (100%)	94.6 / 100	0	Balanced 3-Act
Drift Mode	9 / 21 (42.9%)	92.6 / 100	0	Filters Spikes
Unhinged Engine	21 / 21 (100%)	92.5 / 100	1 (Intentional)	Curveball Drop

Benchmark 2: High-Scale 500-Track Dataset Speed & Retention

Engine Mode	Execution Speed	Throughput Rate	Track Retention	Smoothness
Rise Algorithm	3.18 ms	157,233 tracks/sec	500 / 500 (100%)	99.73 / 100
Drift Mode	4.49 ms	111,359 tracks/sec	138 / 500 (27.6%)	98.66 / 100
Frame Engine	11.02 ms	45,372 tracks/sec	412 / 500 (82.4%)	99.48 / 100
Unhinged Engine	106.22 ms	4,707 tracks/sec	472 / 500 (94.4%)	89.61 / 100

Key Takeaways from Benchmarks

- Ultra-Fast Speed: The Rise Algorithm can sort 500 songs in under 4 milliseconds (over 150,000 songs per second!).
- Near-Perfect Smoothness: Both Rise and Frame engines achieved over 99/100 Smoothness Scores on large datasets.
- Smart Filtering: Drift Mode purposefully filters out extreme tracks to guarantee zero distractions during deep work.

7. How to Run Tunelt Locally

Running Tunelt on your own computer takes just 3 simple steps:

1. Backend Server Setup:

- Open terminal in backend directory: `cd backend`
- Install dependencies: `npm install`
- Add `GEMINI_API_KEY` and `GOOGLE_CLIENT_ID` to your `.env` file
- Run backend server: `npm run dev` (Runs on `http://127.0.0.1:3001`)

2. Frontend Web App Setup:

- Open terminal in frontend directory: `cd frontend`
- Install dependencies: `npm install`
- Run web app: `npm run dev` (Open `http://localhost:3000` in your browser)

3. Running Live Benchmarks:

- Run `npm run evaluate:all` inside the backend folder to see live algorithm scores!

8. Final Summary

Tunelt bridges the gap between raw music collections and curated listening experiences. By combining artificial intelligence with smart mathematical sequencing, Tunelt ensures your music always flows perfectly with whatever you are doing.

Summary & Status

This guide combines project architecture, step-by-step pipeline explanations, simple mode descriptions, and live benchmark readings into a clean 5-page PDF document with zero extra pages.

